

# Disaster Preparedness Education System Website

## 1. Idea Overview

The Disaster Preparedness Education System Website is a web-based platform designed to educate students and communities about disaster awareness, safety procedures, and emergency response strategies. The platform combines structured learning modules, interactive quizzes, and real-time alert systems to improve preparedness and reduce panic during disaster situations.

## 2. Problem Statement

Many communities lack structured disaster education and centralized access to emergency guidelines. During disasters such as earthquakes, floods, or fires, insufficient awareness leads to confusion and delayed response. There is a need for an accessible digital system that educates and prepares individuals effectively.

## 3. Proposed Solution

We propose a centralized educational website that provides disaster-specific learning modules, quizzes for assessment, and emergency alert notifications. The system allows administrators to manage content and update alerts in real time.

## 4. Key Features (MVP)

- User Registration and Secure Login
- Disaster Learning Modules (Earthquake, Flood, Fire, Cyclone, etc.)
- Interactive Quiz with Instant Score
- Emergency Alert Display Section
- Admin Dashboard for Content Management
- Gamified learning

## 5. Tools Used

- VS Code (Development Environment)
- GitHub (Version Control)
- MongoDB Atlas (Database Hosting)
- Render / Netlify (Deployment Platform)
- Figma (UI Design - Optional)

## **6. Languages & Technologies Used**

Frontend: HTML, CSS, JavaScript (Bootstrap)

Backend: Node.js with Express / Python Flask

Database: MongoDB / MySQL

Authentication: JWT & Bcrypt

## **7. Innovation / Unique Value**

- Real-time disaster alerts
- API integration for live weather updates
- AI-based chatbot guidance (future scope)
- Downloadable emergency preparedness guides

## **8. Target Users**

- Students
- Schools & Colleges
- Community Members
- Disaster Management Authorities

## **9. Impact**

- Increases disaster awareness
- Improves emergency response readiness
- Promotes safety culture
- Reduces panic during crisis

## **10. Future Scope**

- Mobile Application
- SMS Alert Integration
- Multi-language Support
- Government Collaboration